Table 9.2: Types and characteristics of tissue systems				
S.No.	**Types/ Characters**	**Epidermal tissue system**	**Ground or fundamental tissue system**	**Vascular or conduction tissue system**
1.	Formation	Forms the outermost covering protoderm	Forms the ground meristem	Forms the procambial bundles
2.	Components	epidermal cells, stomata and epidermal outgrowths	Simple permanent tissues – Parenchyma and Collenchyma	Xylem and Phloem
3.	Functions	Protection of plant body; absorption of water in roots; gas exchange for photosynthesis and respiration; transpiration in shoots	Gives mechanical support to the organs; prepares and stores food in leaf and stem	Conducts water and food; gives mechanical strength

9.4 Epidermal Tissue System

Introduction

Epidermal tissue system is the outer most covering of plants. It is in direct contact with external environment. It consists of epidermis derived from protoderm.Epidermis is derived from two Greek words, namely 'Epi' and 'Derma'. 'Epi' means *upon* and 'Derma' means *skin*. It is made up of single layer of parenchyma cells which are arranged compactly without intercellular spaces. Although epidermis is a continuous outer layer, it is interrupted by stomata in many plants.

Leaf Epidermis

The leaf is generally *dorsiventral*. It has upper and lower epidermis. The epidermis is usually made up of a single layer of cells that are closely packed. Generally the cuticle on the upper epidermis is thicker than that of lower epidermis. The minute openings found on the epidermis are called ***stomata (singular: stoma).*** A stoma is surrounded by a pair of specialised epidermal cells called guard cells. In most dicots and monocots the guard cells are bean-shaped. While in grasses and sedges, the guard cells are dumbbell- shaped. The guard cells contain chloroplasts, whereas the other epidermal cells normally do not have them.

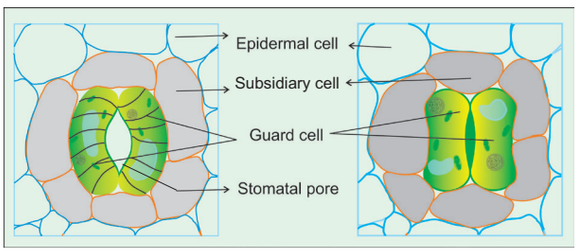


Figure 9.12: (a) Stoma with bean-shaped guard cells. (b) Stoma with dumb-bell shaped guard cells

Check Your Grasp!
In which group of plants the guard cells are dumb-bell shaped?

Grasses and sedges

Subsidiary Cells

Stomata are minute pores surrounded by two guard cells. The stomata occur mainly in the epidermis of leaves. In some plants addition to guard cells, specialised epidermal cells are present which are distinct from other epidermal cells. They are called **Subsidiary cells**. Based on the number and arrangement of subsidiary cells around the guard cells, the various types of stomata are recognised. The guard cells and subsidiary cells help in opening and closing of stomata during gaseous exchange and transpiration.

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